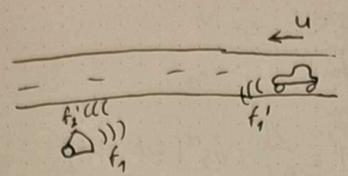
Exercise 1)

1) a) We have that $\Delta t' = \Delta t\left(1-\frac{v}{c}\right)$ at the source

$$\Rightarrow f_{receiven} = f_{source}\left(1+\frac{v_{source}}{c}\right) = 440\left(1+\frac{100}{3,6}\cdot\frac{1}{340}\right) \approx \underline{476\ Hz}$$

b) 

$f_1' = f_1\sqrt{\frac{1+u/c}{1-u/c}}$ f_1' gets transmitted from the car and the police get

$$\Rightarrow f_2' = f_1'\sqrt{\frac{1+u/c}{1-u/c}} = f_1\left(\sqrt{\frac{1+u/c}{1-u/c}}\right)^2 = f_1\left(\frac{1+u/c}{1-u/c}\right) \quad \Big| \ \frac{u}{c}=\gamma, \ \frac{1}{1-\gamma}\approx 1+\gamma$$

$$\approx f_2' = f_1(1+\gamma)(1+\gamma) = f_1(1+2\gamma+\gamma^2) \approx f_1(1+2\gamma) \quad \Big| \ \gamma^2=\frac{u^2}{c^2}\approx 0, \ c>>u$$

$$\Rightarrow \underline{\underline{f_2' = f_1\left(1+2\frac{u}{c}\right)}}$$

for approaching car, swap the + with a - for a car going away.

2) a) $\underline{\zeta_{lsp}(t) = 0,5\cdot 10^{-3}\sin(\omega t)}$ $\ \big|\ \omega = 2\pi f = 2\pi 100$ $\quad \underline{A = -j\hat{\zeta}}$

$$= Re\left|Ae^{j\omega t}\right| = Re\left|A(\cos(\omega t)+j\sin(\omega t))\right| = Re\left|-j\hat{\zeta}(\cos\omega t + j\sin\omega t)\right|$$
$$= \underline{\hat{\zeta}\sin\omega t}$$

b) $v_{lsp}(t) = \frac{d}{dt}\zeta_{lsp}(t) = \underline{\underline{\hat{\zeta}\omega\cos(\omega t)}}$

$$= \frac{d}{dt}Re\left|Ae^{j\omega t}\right| = Re\left|-\omega A\sin\omega t + \omega A j\cos\omega t\right| = \underline{\underline{\hat{\zeta}\omega\cos\omega t}}$$

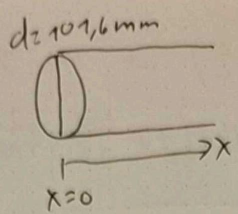
c) $\max V_{lsp} = |\hat{\zeta}\omega| = |2\pi f\hat{\zeta}| = 0,314\ m/s$ for $\omega t = 0, \pi, 2\pi, 3\pi ...$

$$\hat{v} = v(0,0) = V_{lsp}(0) = \hat{\zeta}\omega$$

d) $Z = \rho_0 c = 1,2\cdot 343$ $\quad \max\hat{p} = \hat{v}\cdot Z = 2\pi 100\cdot 0,5\cdot 10^{-3}\cdot 1,2\cdot 343 = \underline{\underline{129\ Pa}}$

e) 1 atm = 101,325 Pa >> 129 Pa Its tiny relative to the atmosphere pressure.

2) f)



$S = \pi/4 \cdot d^2$

We have for a plane wave that

$$p_{real} = A\cos(\omega t - kx)$$
$$v_{x\,real} = \frac{A}{\rho_0 c}\cos(\omega t - kx)$$
$$\Rightarrow p_{real} v_{x\,real} = \frac{A^2}{\rho_0 c}\cos^2(\omega t - kx)$$
$$= \frac{A^2}{2\rho_0 c}\left(1 + \cos 2(\omega t - kx)\right)$$

When integrating over T which is the same as the period of the single-frequency signal, the cos will vanish.

$$I_x = \frac{1}{T}\int_0^T p_{real} v_{real}\,dt = \frac{A^2}{2\rho_0 c} = \frac{1}{\rho_0 c}\left(\frac{A}{\sqrt{2}}\right)^2 = \frac{p_{rms}^2}{\rho_0 c} = \frac{(\zeta^2 \omega^2 \rho_0^2 c^2)}{2\rho_0 c}$$

$W = I_x \cdot S = \underline{0{,}16 \text{ Watt}}$